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1 Introduction

1.1 Scope and Purpose of this document

The purpose of this document is to describe the interface in sufficient detail so that a PMS system can be extended to operate with the Onity locking systems.

1.2 Definitions, Abbreviations, and Acronyms

PMS	Property Management System
IP	Internet Protocol (Ethernet address)
ISO	International Standards Origination (Mag Card track 1,2 format)
Mag Card	Magnetic Strip Card
RS232	Serial port Interface

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2 PMS Connections and Configuration

2.1 Interface: Ethernet TCP/IP

The interface uses Ethernet TCP/IP communications in a typical client-server arrangement. The Onity Encoder is the “server”, and the PMS system can connect as a “client” for using the card-encoding services.

2.2 Interface: RS232

The interface uses RS232 serial. The Onity Encoder is the PMS provider,

2.2.1 RS232 Parameters

The Baud Rate is set on the Onity Encoder. Communication uses 8bits data, 1 or 2 stop bits and no parity.

2.3 Configuration

There are several Onity configuration parameters that are important for PMS connection. These are listed in the following sub-sections.

2.3.1 Encoder ID

The PMS system can encode cards on any of the attached Onity encoders. Each encoder has a unique EncoderID that is used to specify which encoder should perform the card-encoding operation.

2.3.2 Encoder IP

The Onity Encoder IP address must be known in order for the PMS system to connect.

2.3.3 Encoder Port

The Onity Encoder PMS listening port number must be known in order for the PMS to connect. The default value will be 6669

2.3.4 PMS Enable/Disable

The PMS functionality can be enabled or disabled in the Onity system.

2.3.5 PMS Login required/not required

The Onity system can be configured to require a login from the PMS system for any command that encodes a card. The login credentials are supplied on Parameter-11 of the command.

2.3.6 PMS Encoder-Zero mode enable/disable

The Onity system can be configured to allow operation with external encoders. In this mode, when the PMS system needs to encode a card, the Onity system will create a card payload and send that payload back to the PMS system to put onto the card. See the section describing use with 3rd party encoders.

2.3.7 PMS Encoder-Zero Bit-Order

For use with 3rd party encoders (Encoder-Zero mode), the payload is sent back to the PMS system as a sequence of bytes. The encoder will be converting this sequence of bytes into a serial stream of bits to encode onto the card. While converting bytes to a bitstream, there are two ways to interpret the order of

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bits within the byte. This setting provides a way to change the bit-order if the external encoder uses a different interpretation of bit-order in the byte array. This may have to be set using trial and error to find which order works properly on the card to open a lock.
From the Tech Support web interface, go to the System Settings screen and set:

system.config.pms.bitorder_normal
=1 use normal bit-order
=0 use reversed bit-order

2.3.8 Other PMS configuration

Any additional configuration parameters should be left at the default settings:

system.config.pms.loopback_parent = 0
system.config.pms.loopback_child = 0

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3 PMS Protocol Overview

3.1 Special Characters used in the communications

The transmission protocol uses the following ASCII control characters

STX (02H)	Start of message transmission
ENQ (05H)	Communication check (query)
ETX (03H)	End of message transmission
ACK (06H)	Affirmative identification/recognition
NAK (15H)	Negative acknowledgment

3.2 Packet structure

The complete message has the following format:

STX *message* **ETX LRC**

The message itself has two fields:
The command
The command parameters
Every field is delimited by separators. The separator is the ASCII B3h (179 represented as | in this document).

Every packet has to be followed by a longitudinal redundancy check (LRC). The LRC is the character result of the exclusive OR (XOR) of all the message characters (STX not included) plus the ETX character. It has to be calculated with NUL seed (00h).

Instead of calculating the LRC, it is possible to send the ASCII character 13 (RETURN key). The Onity Interface will take it as a correct LRC. This is useful for the PMS communication driver development process, however the final product should include the LRC.

For example, in the sequence STX|CO|0|ETX the LRC is calculated (in HEX) as

(B3)**XOR**(43)**XOR**(4F)**XOR**(B3)**XOR**(30)**XOR**(B3)**XOR**(03)=8C

3.3 Message flow

The control characters ACK and NAK are transmitted by PMS service to indicate affirmative or negative acknowledgment. The Onity Interface answers NAK if it is not already prepared to receive a new message or to indicate that the last message was not correct (bad LRC). If the PMS receives NAK it needs to repeat the message until it receives ACK from the PMS interface. The PMS may send ENQ to know if the Onity Interface is free to receive a request message.

See the following example of communication below:

PMS Onity Interface

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STX *message* **ETX LRC**

NAK

ENQ

ACK

STX *message* **ETX LRC**

ACK

•

•

•

STX operation result **ETX LRC**

The PMS needs to perform the following operations:

- Calculate and send the LRC of the characters following STX including the ETX (The LRC can be simulated by the ASCII character #13).
- Wait until the acknowledgment (ACK) of the previous task before sending a new one.
- If there is no acknowledgment for 2 seconds, the PMS has to decide what to do because the link with the Onity Interface has been cut.
- Show in the corresponding monitor the messages corresponding to every operation in order to guide the operation.

The Onity Interface software performs the following operations:

- Checks the LRC to determine the integrity of the message received.
- Replies ACK or NAK before executing the received message.
- Replies NAK if the LRC is not correct.
- Execute the task until done, or until the PMS aborts it.
- The Onity encoders have messages displayed on-screen to prompt the user to insert cards at the proper time.
- When done, send the result packet back to the PMS

3.4 Authentication

The system can be configured to require the PMS to supply login credentials with each message. The username and password must be packed into the PMS message, in parameter-11, “Data of operator encoding the card”.

If this string has a colon “:” in it, it will be interpreted as “username:password”. If the string doesn’t have a colon, the string value should be used as both the username and password.

Example:

Parameter-11	Username	Password
pmssystem	Pmssystem	pmssystem
pms:system	Pms	system

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4 Commands

This section lists all the supported commands for the Onity PMS interface. For a description of the parameters of these commands, refer to the several sections that follow.

4.1 [CNx] Check-in a Guest and encode a card

To encode a new guest key card and *x* copies minus one (total of x cards). This card will automatically cancel the previous guest's key card. When using the virtual encoder (encoder zero) mode, x should be set to 1.

4.2 [CO] Check-out a Guest

Check-out of a guest. All guest cards of the corresponding room will be canceled.

4.3 [CCx] Create Copy of a Guest Card

To encode copy(s) of a valid new key card. *x* is the number of copy(s) that will be made. When using the virtual encoder (encoder zero) mode, x should be set to 1.

4.4 [CA] Make a Single-Opening Card

To encode a single opening card. The single opening cards are only valid for that particular guest room lock. They are automatically canceled after check out of the guest or after having been used the first time.
Note, OnPortal will accept CA1 in addition to CA.

4.5 [LT] Read a Guest Card

Read and interpret data on track 3 according the Onity standard.

4.6 [P1] Encode Track-1 ISO data

Encode information on track 1 of the magnetic card according to the ISO standard for track 1. This is only available when using the motorized encoder module, not the manual insertion module.

4.7 [P2] Encode Track-2 ISO data

Encode information on track 2 of the magnetic card according to the ISO standard for track 1. This is only available when using the motorized encoder module, not the manual insertion module.

4.8 [L1] Read Track-1 ISO data

Read the information on track 1, according to the ISO standard for track 1. This is only available when using the motorized encoder module, not the manual insertion module.

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4.9 [L2] Read Track-2 ISO data

Read the information on track 1, according to the ISO standard for track 1. This is only available when using the motorized encoder module, not the manual insertion module.

4.10 [EX] Reset Encoder

Resets the encoder. Ejects card (if any) out of the encoder and aborts the last non completed operation.

4.11 [VS] PMS Protocol Version

Requests the PMS Protocol Version. This command is not implemented on Onity Encoders nor on HT28 nor other older implementations of the protocol. If not implemented, and error response is expected. If implemented, then a response with a version is expected.

STX|VS|ETXLRC

STX|VS|2.0**|ETXLRC**

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5 Parameters for encoding commands (CNx, CCx, CA)

There are up to THIRTEEN (FOURTEEN with a KeyID in an updated version) parameters for the card-encoding commands (after the command field “CNx”. The message needs to contain at least the first three parameters. The others are optional. If some parameters are omitted, their (empty) fields have to be put in the message with field separators until the last non omitted parameter.

- Command field (CNx, CCx, or CA)
- P1: EncoderID: identifier of the encoder to encode the card.
- P2: Retain or eject the card.
- P3: Room assigned as no. 1. (primary room)
- P4: Room assigned as no 2.
- P5: Room assigned as no 3.
- P6: Room assigned as no 4.
- P7: Authorizations assigned to guest.
- P8: Authorizations denied to guest.
- P9: Initial (start) date of the card.
- P10: Expiration date of the card.
- P11: Operator encoding the card.
- P12: Information encoded on track 1.
- P13: Information encoded on track 2.
- P14: KeyID information. (Onportal Only).

5.1 *x*: The number of card(s) to be encoded

The CNx and CCx commands have this value in the command field.
Several cards for the same room can be issued with one single request.
The variable *x* can be an integer from 1 to 9 (x can also be left blank and will be read as “1”). During the encoding operation, the Onity Encoder will prompt the operator to insert each card, one at a time.
For instance:

Using CN4 command will issue 1 new guest card + 3 copies.
Using CC3 command will issue 3 guest card copies.
Using CN1 will issue only a new guest card.
Using CN will issue only a new guest card.

5.2 Parameter 1: EncoderID

The PMS system can specify an number from 1…99 that identifies which Onity Encoder in the system should do the card encoding. These numbers are set on the config screen of each encoder. When this number is zero (0), it has special meaning. See the section for Encoder-Zero mode.

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5.3 Parameter 2 : Ejection or retention of the card

When finishing the encoding operation, the card can remain in the encoder for successive operations or it can be ejected.

E: Ejection (typical use)
R: Retention

5.4 Parameters 3, 4, 5 and 6: Rooms

This field specifies which rooms that can be opened by the card. Most of the time a card is encoded for a single room, but sometimes a guest card will be made that can open 2, 3, or 4 rooms. When fields are unused, the empty field with separator character must still be included in the message.

- NOTE: These room name strings must exactly match the room names that have been configured in the Onity system for this hotel.
- NOTE: When the encoder has the motorized encoding deck, it can encode up to 4 rooms, but when using the manual insertion deck, it can only encode up to 3 rooms per card.
- NOTE: The command CA (single opening card) only works for the **first,** or principle room, so parameters 4, 5 & 6 are meaningless.

5.5 Parameter 7 : Assigned authorizations (Max. 8 authorizations)

The Onity locking system will define up to 8 authorizations that can be assigned to the guest cards. These will typically correspond to hotel amenities such as the Pool or the Parking deck. These authorizations are identified in this protocol with numbers 1 through 8.
Specify the numbers of the authorization assigned to the guest.

Warning: The number given to each authorization and its description can differ in each hotel. Please consult Onity for the number and description of each authorization.

5.6 Parameter 8: Denied Authorizations (Max. 8 authorizations)

Specify the numbers of the authorization denied to the guest. These Denied Authorizations will remove authorization the guest would have been given by default. For example, if room 101 is given the Pool authorization (1) by default, specifying 1 in this parameter will remove that authorization so it is not given to the guest.

5.7 Parameter 9: Initial date

This field specifies the guest checkin date-time, or date-time when the card will start working in the room lock.
There are eight characters total, HHDDMMYY:

HH Hour (00 to 23)
DD Day (01 to 31)
MM Month (01 to 12)
YY Year (11=2011, 12=2012, etc)

For example:

"12100712" Indicates a start date of 12-noon, 10th of July, 2012

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Note: Complete omission of this field means that the card will be valid immediately after encoding it.

Note: If motorized encoders are used and four rooms are required on a key card, the initial date must be omitted.

5.8 Parameter 10: Expiration date

Date on which the card will become invalid. This command has the same format as the initial date. Omitting this field indicates that the card has no time limit. It will be canceled by the next new guest card. The longest expiration date goes up to the 31st of December of the next year.
Important: The Onity system allows the encoding of an initial date only if an expiration date is also encoded. If the encoding of a card with only initial date is requested, the Onity Interface will answer with a syntax error message (ES). Please also note that the date MUST be in “European” format. (HH DD MM YY)

5.9 Parameter 11: Operator encoding the card (Max. 20 characters)

The audit trail of the Onity system will register the encoding operation of this card with the name of the operator written in this field. If you can extract the name of the user who is logged into this terminal it is a good idea to insert it into this field. Most properties management staff would like this to display the initials of the user. Complete omission of this field means that the operation will be registered with the name "PMS" and in most cases is unacceptable to the management staff.

Optional Password Validation

With this feature enabled, the data included in parameter 11 of the interface data packet will be validated against the passwords in the Operator table of the Onity software. If the password is valid, the key encoding is allowed. If the password is not valid, an error is returned.

See the section on Authentication for more information.

5.10 Parameter 12: Track 1 data

This command is supported for HT24W/HT28 and OnPoint. HT22 does not support this command. Specify the data to be encoded on Track 1, but do NOT include the start/end sentinels. The sentinels will be automatically added by the Onity System.

5.11 Parameter 13: Track 2 data

This command is supported for HT24W/HT28 and OnPoint. HT22 does not support this command. Specify the data to be encoded on Track 2, but do NOT include the start/end sentinels. The sentinels will be automatically added by the Onity System.

5.12 Parameter 14: Key ID

The KeyID field would be used for properties with DirectKey Software as a Service enabled. The purpose of this field is to designate a unique digital Key ID for which to send this card’s payload to.

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The KeyID field can be used for either the Key.Name (Usualy an Email Address of the Guest phone), or an explicit Key Serial Number from the Onity Core. If a physical Encoder# is used on a command with a KeyID a physical card will be encoded in addition to the DirectKey.

5.13 Example Command

STX|CN1|2|E|RM101||||2468|1357||13060205|pms:xyz||||ETXLRC

The above line is an example of the data packet received from the PMS system to encode a new card for room RM101. This message is broken down as follows:
STX – the Start of Text command
Command – ‘CN1’ – New Card command, single card
Parameter 1 – ‘2’ – Use Encoder 2
Parameter 2 – ‘E’ – Eject card (for use with motorized encoders)
Parameter 3 – ‘RM101’ – Room RM101
Parameters 4, 5, and 6 – Used for multiple room cards, empty in this example
Parameter 7 – ‘2468’ – Assigned Authorizations
Parameter 8 – ‘1357’ – Denied Authorizations
Parameter 9 – ‘ ’ – Start date and time; empty means valid starting now.
Parameter 10 – ‘13060212’ – Expiration (HHDDMMYY): 1pm, 6 Feb, 2012
Parameter 11 – ‘pms:xyz’ – Userid “pms”, Password “xyz”
Parameter 12 – ‘ ’ – Track 1 Data
Parameter 13 – ‘ ’ – Track 2 Data
Parameter 14 – ‘ ‘ – KeyID to later re-identify the Generated Payload data.
ETX – the End of Text command

And the response would be:

STX|CN1|2|7B887824*|ETXLRC
* When using the OnPortal system with Mifare Cards, one can configure the system to return the UID of the card that was encoded as an additional parameter of the card command response.

For Encoder 0 response, see Chapter on “Third Party Encoders”.

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6 Parameters for command CO (Checkout)

This command has two parameters:

Encoder number (The encoder number for this operation is always 0, as an encoder is not used.)

Room Name

Example:
Let us suppose that we want to cancel the card of room RM101 (check-out of the corresponding guest).
The PMS has to send the message:

STX|CO|0|RM101|**ETXLRC**

If the operation has ended without errors the Onity Interface will send the following message to the PMS:
STX|CO|0|**ETXLRC**

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7 Parameters for command EX (Encoder Reset)

This command has two parameters:

- Number of encoder
- frontal ejection, retention or rear ejection

Example:
Let us suppose that we want to reset encoder 1. The PMS has to send the message:

STX|EX|1|E|**ETXLRC**

If the operation has ended without errors the Onity Interface will send the following message to the PMS:

STX|EX|1|**ETXLRC**

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8 Parameters for the command LT (Read a Guest Card)

The PMS can recognize every card of the hotel, but will only identify a valid guest card. This command is used to discover why a card may not be working in a particular room or may not operate a particular authorization.
This command will cause one of the following 4 messages to be returned by the Onity System:

1. Card recognized and identified
2. Card recognized, but not identified
3. Card not recognized
4. Error message (see the section on error messages for a complete list)

8.1 Read Card Request (LT)

Ex: STX|LT|1|E |ETXLRC

#	Parameter Name	Example
0	Read Card Command	LT
1	Encoder ID	3
2	Card Eject	E

8.2 Read Card Response (Recognized Guest Card)

STX|LT|1|101|102|||CI|2|126||12150693|PMS|7B887824**ETXLRC**

#	Parameter Name	Example
0	Read Card Response (no error)	LT
1	Encoder ID	3
2	Room 1	101
3	Room 2	102
4	Room 3	
5	Room 4	
6	**Card Status**	CI
7	Copy # of primary door	2
8	Authorizations	
9	Start Date [HHDDMMYY]	
10	End Date [HHDDMMYY]	12150693
11	Operator	PMS
12	Mifare Card UID*	7B887824

***Mifare Card UID is a unique ID embedded in each Mifare card. When using the OnPortal system with the Onportal (PCSC) encoder, the system can be configured to return this value when a card is read.**

8.2.1 Card Status Definitions

Code	Description

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CI	Card with valid primary room.
CO	Card where the primary room has been canceled by check out.

8.3 Read Card Response (NON-Recognized or NON-Guest Card)

Example: **STX**|LT|3|LD|**ETXLRC**

#	Parameter Name	Example
0	Read Card Response (no error)	LT
1	Encoder ID	3
6	Card Status	LD

8.3.1 Card Status Definitions

Code	Description
LC	Guest card NOT valid
LM	Master card or special card.(programming, canceling or blocking card)
LR	Spare card (backup card).
LS	Diagnostic card
LD	Unidentified card (Belongs to another hotel or has been erased in the locking plan).

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9 Parameters for commands P1, P2 (Encode ISO Track 1/2)

The Onity interface translates the ASCII characters into ISO symbols for track 1, 2 or 3. It also calculates the parity of the message in such a way that it respects the ISO norms for track 1, 2 or 3.

IMPORTANT: Read the ISO specification for track 1, 2 and 3 in for the following information(refer Section 13):

- Allowed symbols for each track
- Maximum number of characters for each track
- Characters for start, end, and separator for each track.

9.1 Encode ISO Request(P1 or P2)

STX|P1|1|R|%SQUASH CLUB MEMBER?|**ETXLRC**
STX|P2|1|R|;23456?|**ETXLRC**

#	Parameter Name	Example
0	Encode ISO Command	P1 (Track1) or P2 (Track 2)
1	Encoder ID	3
2	Card Eject	R
3	TrackData	%SQUASH CLUB MEMBER? (Track 1) or ;23456? (Track 2)

9.2 Encode ISO Response (Success)

STX|P1|1|ETXLRC

#	Parameter Name	Example
0	Encode ISO Response (no error)	P1 (Track1) or P2 (Track 2)
1	Encoder ID	3

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10 Parameters for commands L1 and L2 (Read ISO track 1/2)

If we want to read the encoded card of the previous Encode ISO Track, eject it at the end of the operation and use below commands.

10.1 Read ISO Request(L1 or L2)

STX|L1|1|E|**ETXLRC**
STX|L2|1|E|**ETXLRC**

#	Parameter Name	Example
0	Read ISO Command	L1 (Track1) or L2 (Track 2)
1	Encoder ID	3
2	Card Eject	E

10.2 Read ISO Response (Success)

STX|L1|1|%SQUASH CLUB MEMBER?|**ETXLRC**

#	Parameter Name	Example
0	Read ISO Response (no error)	L1 (Track1) or L2 (Track 2)
1	Encoder ID	3
2	Track Data	%SQUASH CLUB MEMBER?

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11 Error Messages

If the operation requested by the PMS could not be carried out correctly, the Onity system will return an error message. The error message is followed by the number of the corresponding encoder.

The PMS must display the error message on the appropriate PMS terminal screen and repeat the transmitted message if it was not a fatal error. For example, if the card was inserted backwards, simply pause, alert the operator, and reissue the same command.

ERROR MESSAGES:

EA Requested copy corresponds to a room that has been checked-out. This message also appears when requesting a fifth one shot card, because a maximum of 4 one shot cards per each new guest can be encoded.
EB Onity encoder is busy.
ED Card not inserted or jammed in the encoder.
EE Operation Canceled. The requested function was canceled by the Onity program. This can occur when shutting down the PMS communications link for special operations.
EF Magnetic format error. The magnetic strip may be damaged.
EO Requested room data being processed by another station.
EP Card inserted incorrectly. The card was inserted backwards or does not have a magnetic stripe.
ES Syntax error. The message envelope or format is not correct (unknown command, invalid parameters, invalid characters, or too many characters to write according to the ISO norms, incorrect date or date format, etc.).
ET Card jammed in encoder. The card does not have the proper physical dimensions and is stuck in the encoder, or a foreign object may have been inserted into the encoder.
EZ Encoder zero is not supported.
NC No communication. The corresponding encoder does not answer. Failure in the communications, the encoder is not turned on or plugged in, or the encoder does not exist.
OS Room out of service.
OV Overflow (busy). The encoder or operator has not performed the previous task.
PE Password Error. If using the password validation option, an invalid password in the operator name field, parameter 11 will generate this error.
TD Unknown room. The room number requested by the PMS system does not exist in the Onity system.

Example:
The PMS requests the encoding of a new guest card for room 101 in encoder number 2. All others parameters use the system defaults.

The PMS sends to the Encoder:

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STX|CN|2|E|101|**ETXLRC**

If the card has been inserted backwards the Onity interface answers:

STX|EP|2|**ETXLRC**

The PMS displays the error on the screen of the appropriate terminal, and then sends through the command again.

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13 ISO specifications for tracks

Track #	**Maximum # of characters**	**Allowed symbols**
1	Allows the encoding of 79 characters, out of which 76 characters are meant for message. Other characters are 2 control characters, and the 1 character LRC • start sentinel character '%' • end sentinel character '?' • redundancy check (LRC) character	There are 64 valid characters **SP ! " # $ % & ' ( ) * + , - . /** **0 1 2 3 4 5 6 7 8 9 : ; < = > ?** **@ A B C D E F G H I J K L M N O** **P Q R S T U V W X Y Z [\ ] ^ _** **Note:** Note that only capital letters are available.
2	Allows the encoding of 40 characters, out of which 37 characters are meant for message. Other characters are 2 control characters, and the 1 character LRC • start sentinel character ';' • end sentinel character '?' • redundancy check (LRC) character	There are 16 valid characters **0 1 2 3 4 5 6 7 8 9 : ; < = > ?**

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